

Module 2

Transient Analysis of First and Second-Order Circuits

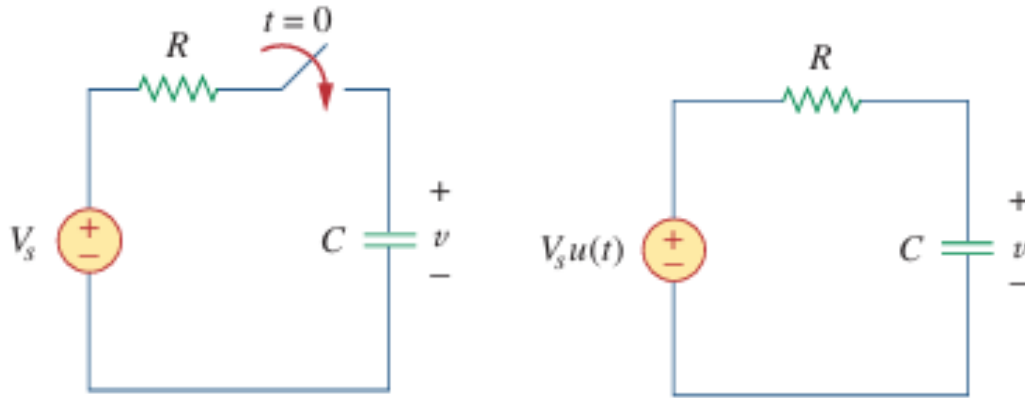
Dr. Elizabeth Rufus

Professor, School of Electronics Engineering

Step Response of RC circuit

The step response of a circuit is its behavior when the excitation is the step function, which may be a voltage or a current source.

When the dc source of an RC circuit is suddenly applied, the voltage or current source can be modeled as a step function.



Initial voltage on the capacitor is V_0

$$v(0^-) = v(0^+) = V_0$$

Where $v(0^-)$ is the voltage across the capacitor just before switching and $v(0^+)$ is its voltage immediately after switching.

Applying KCL;

$$C \frac{dv}{dt} + \frac{v - V_s u(t)}{R} = 0$$

$$\frac{dv}{dt} + \frac{v}{RC} = \frac{V_s}{RC} u(t)$$

Where v is the voltage across the capacitor. For $t > 0$,

$$\frac{dv}{dt} + \frac{v}{RC} = \frac{V_s}{RC}$$

Step Response of RC circuit

$$\frac{dv}{dt} + \frac{v}{RC} = \frac{V_s}{RC}$$

Rearranging terms gives,

$$\frac{dv}{dt} = -\frac{v - V_s}{RC}$$

$$\frac{dv}{v - V_s} = -\frac{dt}{RC}$$

Integrating both sides and introducing the initial conditions,

$$\ln(v - V_s) \Big|_{V_0}^{v(t)} = -\frac{t}{RC} \Big|_0^t$$

$$\ln(v(t) - V_s) - \ln(V_0 - V_s) = -\frac{t}{RC} + 0$$

$$\ln \frac{v - V_s}{V_0 - V_s} = -\frac{t}{RC}$$

Step Response of RC circuit

Taking the exponential of both sides,

$$\frac{v - V_s}{V_0 - V_s} = e^{-t/\tau}, \quad \tau = RC$$
$$v - V_s = (V_0 - V_s)e^{-t/\tau}$$

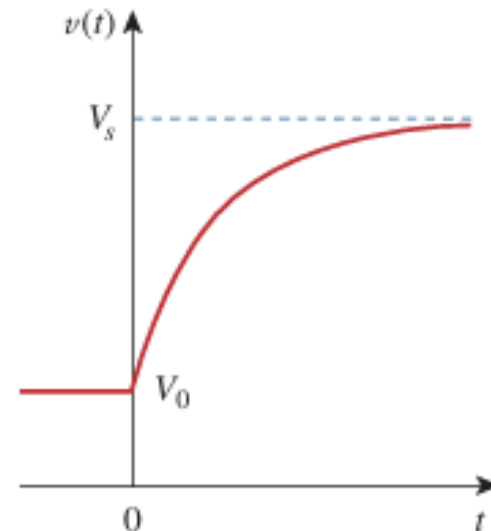
$$v(t) = V_s + (V_0 - V_s)e^{-t/\tau}, \quad t > 0$$

Thus,

$$v(t) = \begin{cases} V_0, & t < 0 \\ V_s + (V_0 - V_s)e^{-t/\tau}, & t > 0 \end{cases}$$

Assuming that $V_s > V_0$ a plot of $v(t)$ is shown:

This is known as the **complete response** (or total response) of the RC circuit to a sudden application of a dc voltage source, **assuming the capacitor is initially charged**.

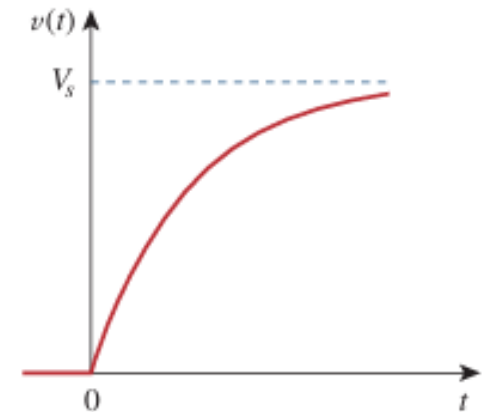


Step Response of RC circuit

If $V_o = 0$,

$$v(t) = \begin{cases} 0, & t < 0 \\ V_s(1 - e^{-t/\tau}), & t > 0 \end{cases}$$

this can be written alternatively as, $v(t) = V_s(1 - e^{-t/\tau})u(t)$

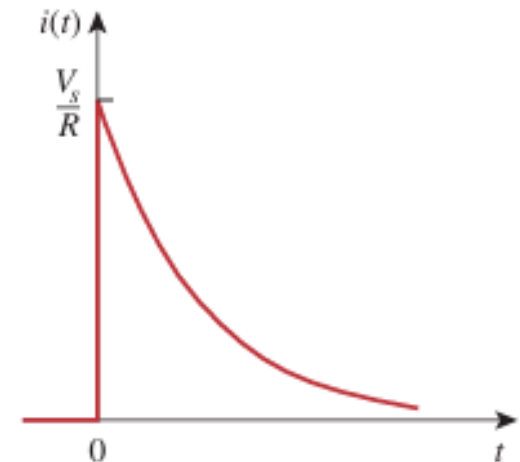


This is the **complete step response** of the RC circuit when the **capacitor is initially uncharged**.

The current through the capacitor is,

$$i(t) = C \frac{dv}{dt} = \frac{C}{\tau} V_s e^{-t/\tau}, \quad \tau = RC, \quad t > 0$$

$$i(t) = \frac{V_s}{R} e^{-t/\tau} u(t)$$



Complete response = natural response + forced response
stored energy independent source

$$v = v_n + v_f$$

(based on the sources of response)

$$v_n = V_o e^{-t/\tau}$$

$$v_f = V_s(1 - e^{-t/\tau})$$

Step Response of RC circuit

Another way of looking at the complete response is,

$$\text{Complete response} = \underset{\text{temporary part}}{\text{transient response}} + \underset{\text{permanent part}}{\text{steady-state response}}$$

(based on the permanency of response)

$$v(t) = V_s + (V_0 - V_s)e^{-t/\tau}, \quad t > 0$$

$$v = v_t + v_{ss}$$

$$v_t = (V_0 - V_s)e^{-t/\tau} \quad v_{ss} = V_s$$

The transient response v_t is temporary; it is the portion of the complete response that decays to zero as time approaches infinity.

The steady-state response V_{ss} is the portion of the complete response that remains after the transient response has died out.

In general,

$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$$

where $v(0)$ is the initial voltage at $t=0^+$ and $v(\infty)$ is the final or steady state value.

Step Response of RC circuit

If the switch changes position at time $t = t_0$ instead of $t = 0$ there is a time delay in the response so that the equation becomes,

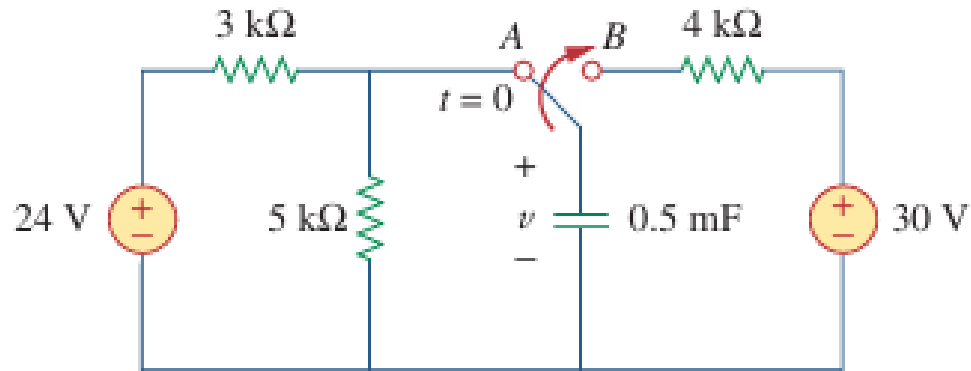
$$v(t) = v(\infty) + [v(t_0) - v(\infty)]e^{-(t-t_0)/\tau} \quad \text{Where } v(t_0) \text{ is the initial value at } t = t_0^+$$

Instead of

$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$$

Exercise

1. The switch in the circuit below has been in position A for a long time. At $t = 0$ the switch moves to B. Determine $v(t)$ for $t > 0$ and calculate its value at $t = 1$ s and 4 s.



For the time $t < 0$, switch is at position A, so v_c will be the same as drop across $5 \text{ k}\Omega$ resistor.

$$v(0^-) = \frac{5}{5 + 3}(24) = 15 \text{ V}$$

$$v(0) = v(0^-) = v(0^+) = 15 \text{ V}$$

For the time $t > 0$, switch is at position B, $R_{\text{th}} = 4 \text{ k}\Omega$

$$\tau = R_{\text{Th}}C = 4 \times 10^3 \times 0.5 \times 10^{-3} = 2 \text{ s} \quad \text{and} \quad v(\infty) = 30 \text{ V}.$$

Exercise

$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$$

$$= 30 + (15 - 30)e^{-t/2} = (30 - 15e^{-0.5t}) \text{ V}$$

At $t = 1$,

$$v(1) = 30 - 15e^{-0.5} = 20.9 \text{ V}$$

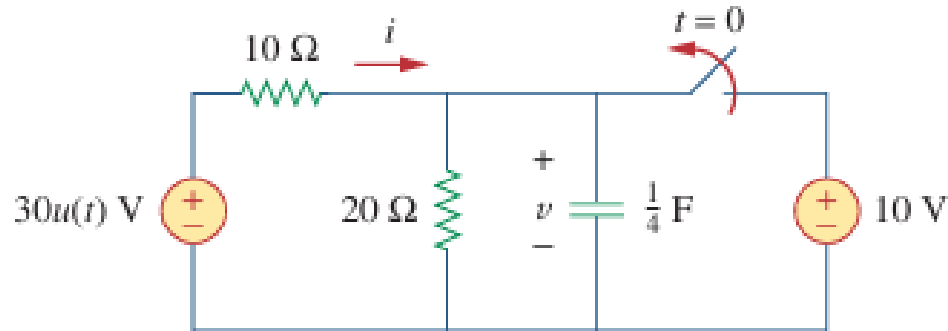
At $t = 4$,

$$v(4) = 30 - 15e^{-2} = 27.97 \text{ V}$$

Exercise

In the circuit given below, the switch has been closed for a long time and is opened at $t = 0$.

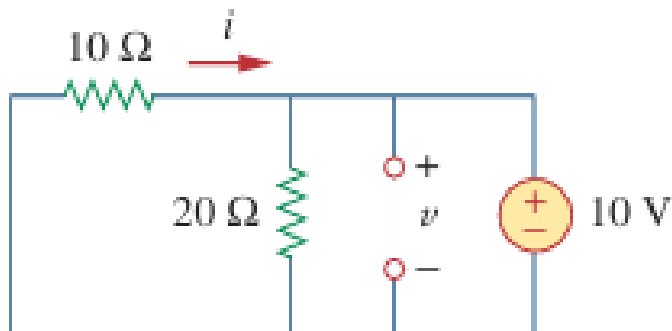
Find i and v for all time.



By definition of the unit step function,

$$30u(t) = \begin{cases} 0, & t < 0 \\ 30, & t > 0 \end{cases}$$

For $t < 0$, the switch is closed and so that the voltage source is replaced by a short circuit and should be regarded as contributing nothing to v .

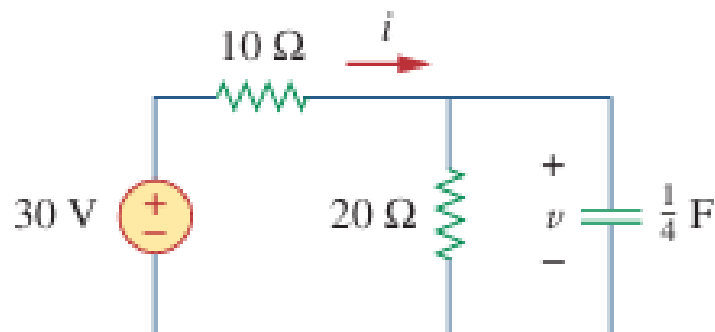


$$v = 10 \text{ V}, \quad i = -\frac{v}{10} = -1 \text{ A}$$

$$v(0) = v(0^-) = 10 \text{ V}$$

Exercise

For $t > 0$, the switch is opened and the 10-V voltage source is disconnected from the circuit.



$$v(\infty) = \frac{20}{20 + 10}(30) = 20 \text{ V}$$

$$R_{\text{Th}} = 10 \parallel 20 = \frac{10 \times 20}{30} = \frac{20}{3} \Omega$$

$$\tau = R_{\text{Th}} C = \frac{20}{3} \cdot \frac{1}{4} = \frac{5}{3} \text{ s}$$

$$\begin{aligned} v(t) &= v(\infty) + [v(0) - v(\infty)]e^{-t/\tau} \\ &= 20 + (10 - 20)e^{-(3/5)t} = (20 - 10e^{-0.6t}) \text{ V} \end{aligned}$$

$$\begin{aligned} i &= \frac{v}{20} + C \frac{dv}{dt} \\ &= 1 - 0.5e^{-0.6t} + 0.25(-0.6)(-10)e^{-0.6t} = (1 + e^{-0.6t}) \text{ A} \end{aligned}$$

$$v = \begin{cases} 10 \text{ V}, & t < 0 \\ (20 - 10e^{-0.6t}) \text{ V}, & t \geq 0 \end{cases}$$

$$i = \begin{cases} -1 \text{ A}, & t < 0 \\ (1 + e^{-0.6t}) \text{ A}, & t > 0 \end{cases}$$

Next class: Step Response of an RL circuit